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Schroeder et al.

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(54) **ELECTRIC VEHICLE CHARGER WITH
AUTOMATED GRID-MANAGEMENT AND
VEHICLE-TO-HOME ISOLATION/BACKUP**

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See application file for complete search history.

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U.S.C. 154(b) by 0 days.

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H02J 7/00 (2006.01)

H02J 13/00 (2006.01)

(52) **U.S. Cl.**

CPC **B60L 53/60** (2019.02); **H02J 7/0029**
(2013.01); **H02J 13/00002** (2020.01)

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B60L 55/00; H02J 7/0029; H02J
13/00002; H02J 13/001; H02J 13/322;

Primary Examiner — Julian D Huffman

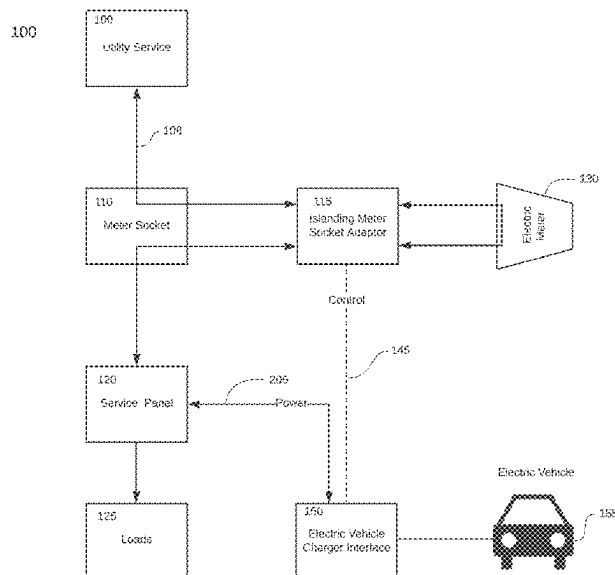
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(57) **ABSTRACT**

An interface to an electric vehicle or comparable energy storage system, paired with a utility electric power meter or meter collar adaptor with embedded control and whole-site disconnect switch that, when operated together, enables the user to: isolate the site from the grid during a power outage and use the vehicle or energy storage system to provide power to the site; manually reclose the disconnect switch to the grid only when safe operating conditions are met; allow a remote grid-operator to manage charging of the vehicle; and prevent the utility service from being overloaded when providing power to the residence and the meter adaptor loads simultaneously.

17 Claims, 23 Drawing Sheets



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FIG. 1

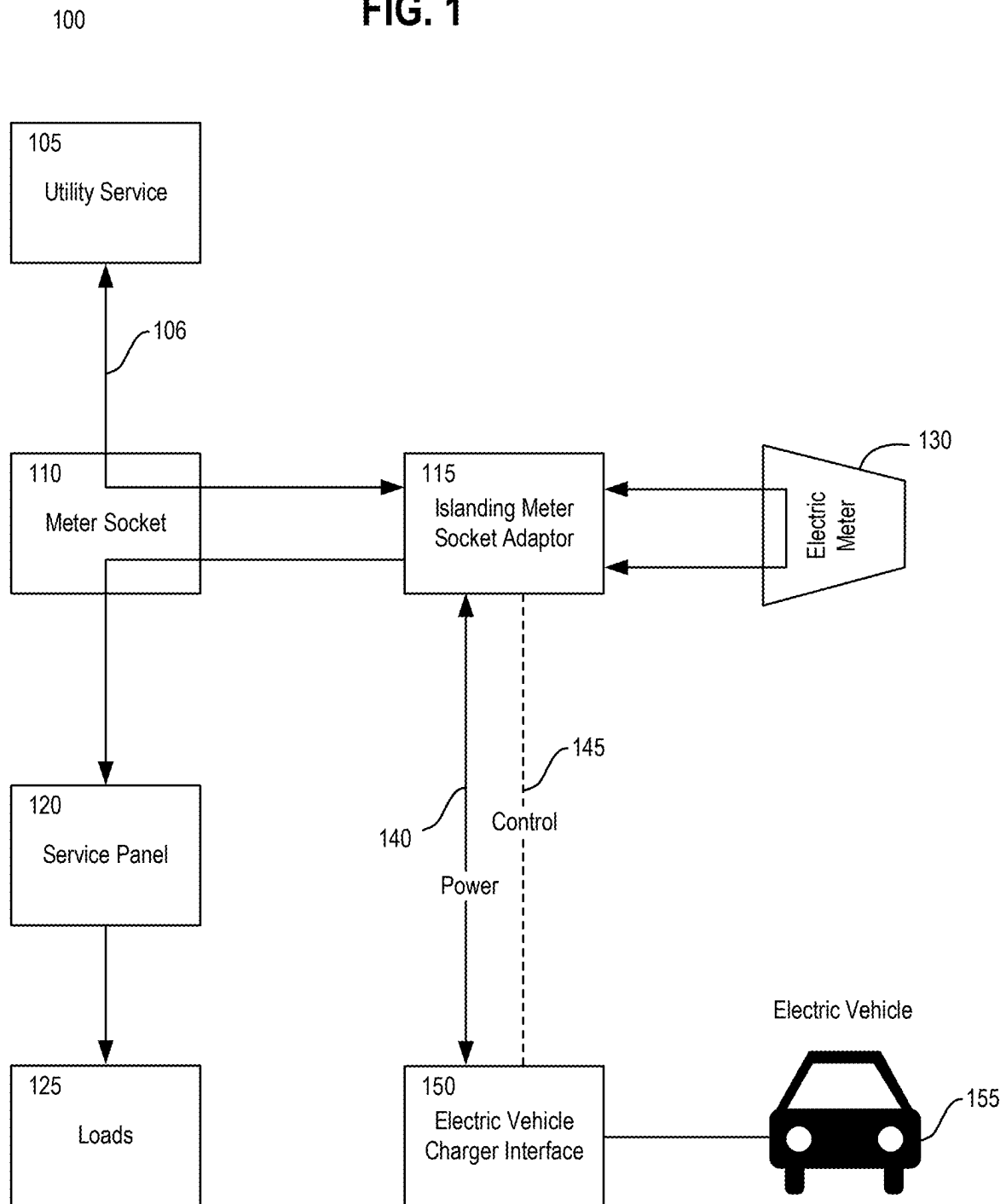


FIG. 2

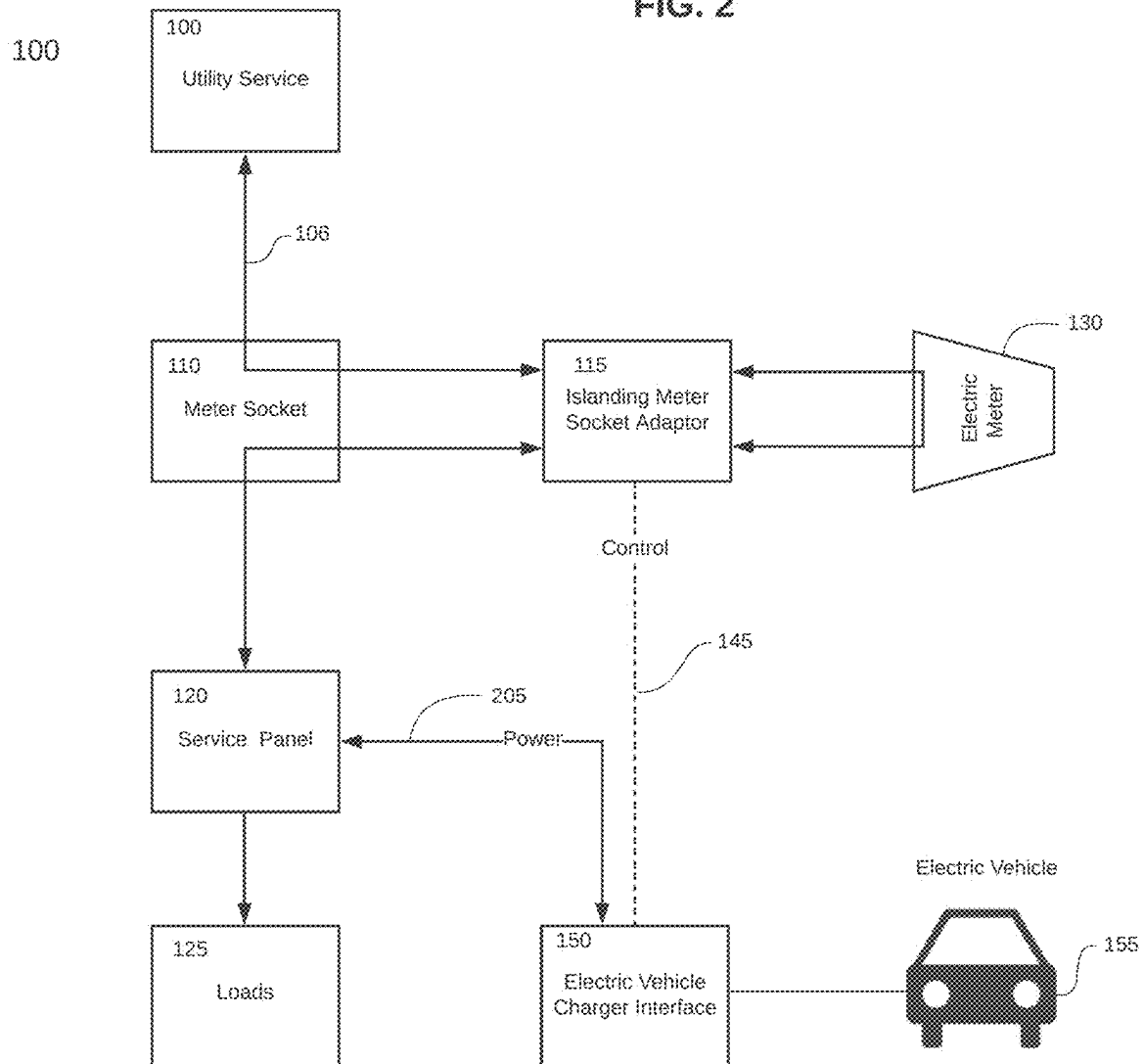
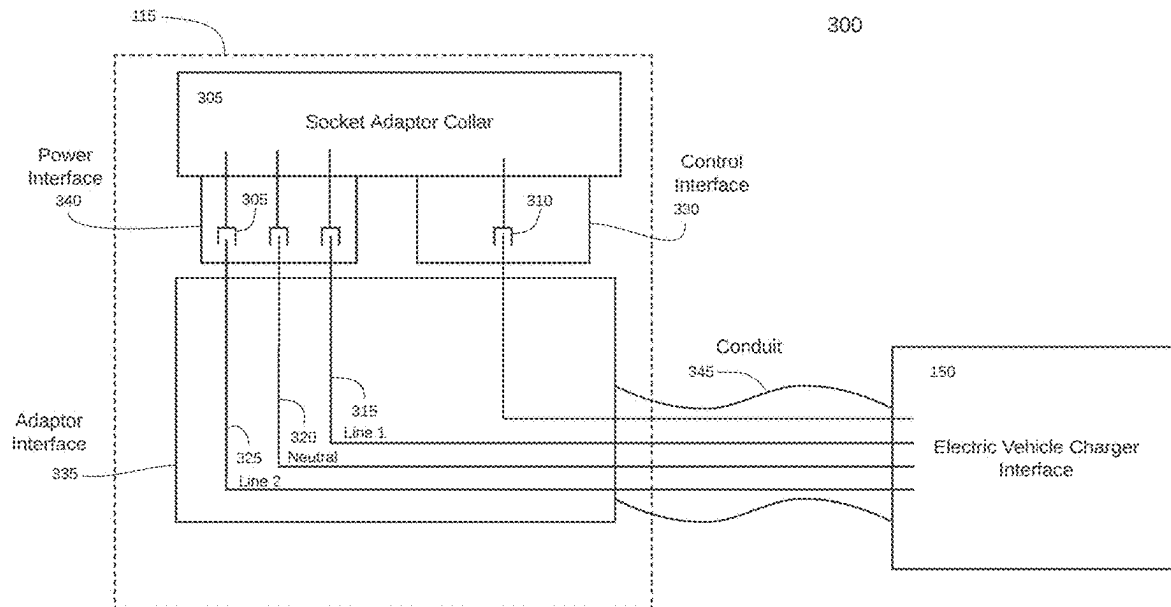


FIG. 3



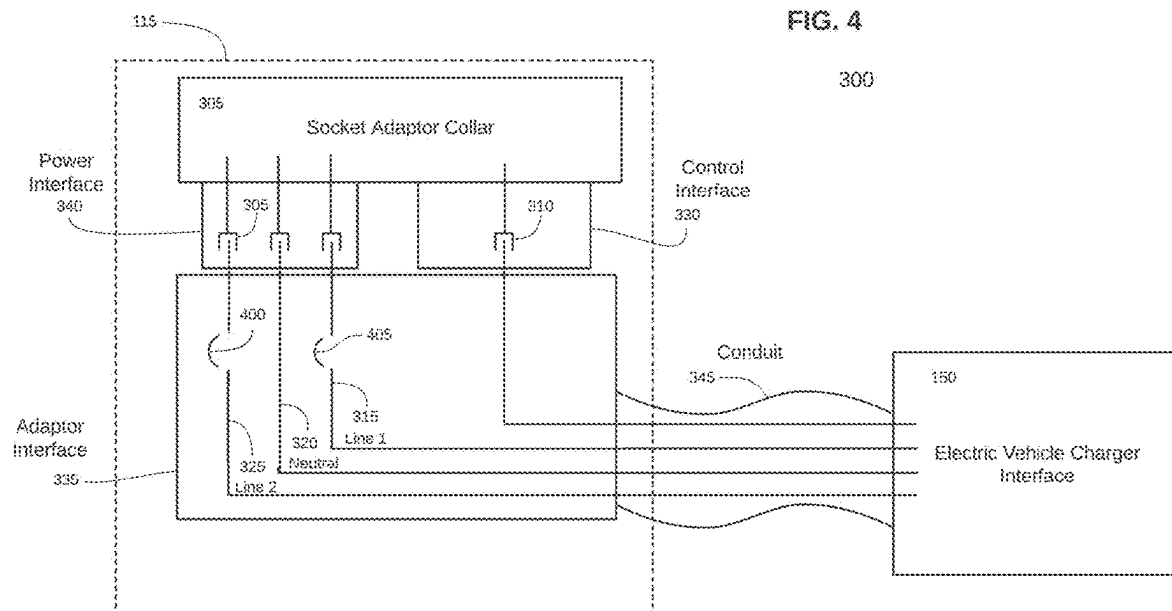


FIG. 5

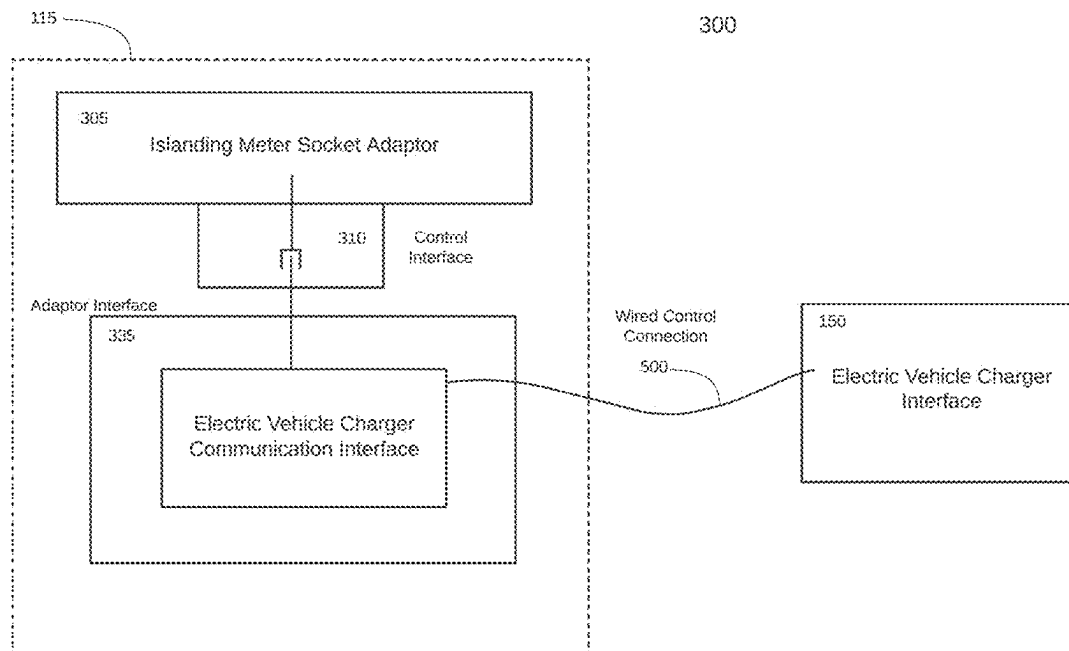


FIG. 6

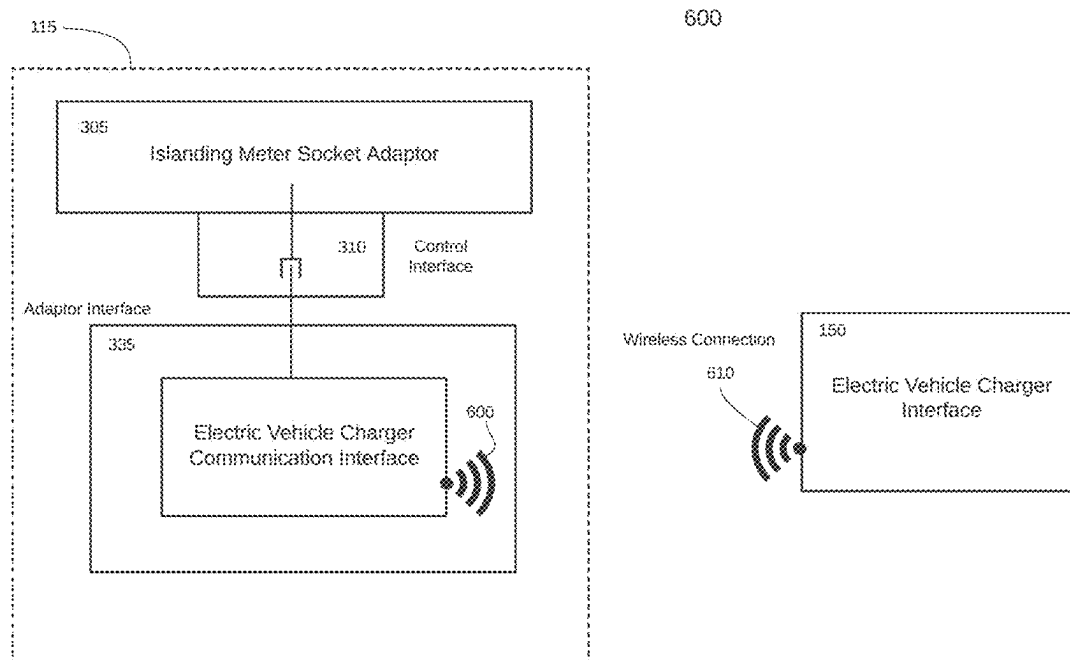


FIG. 7

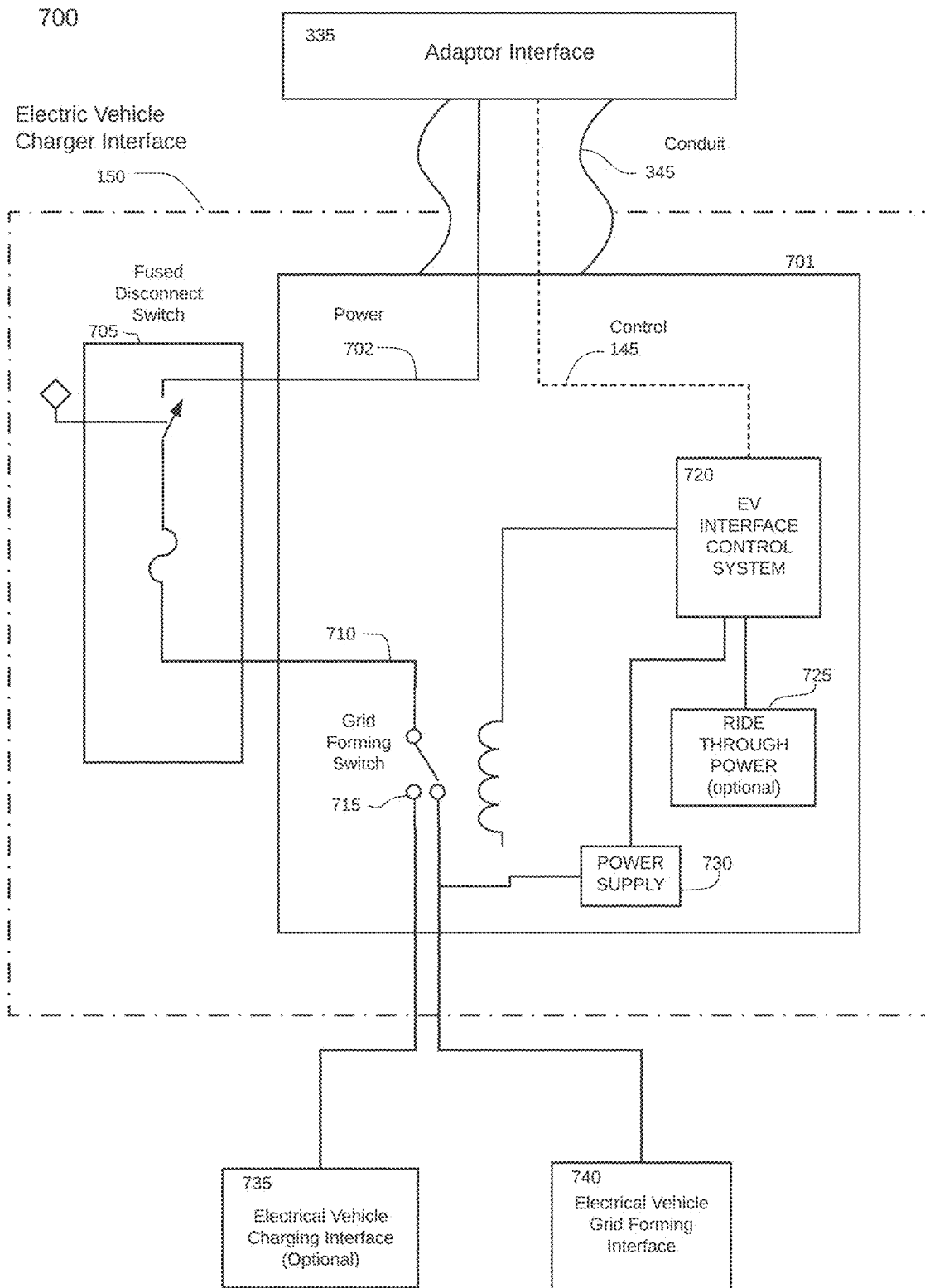
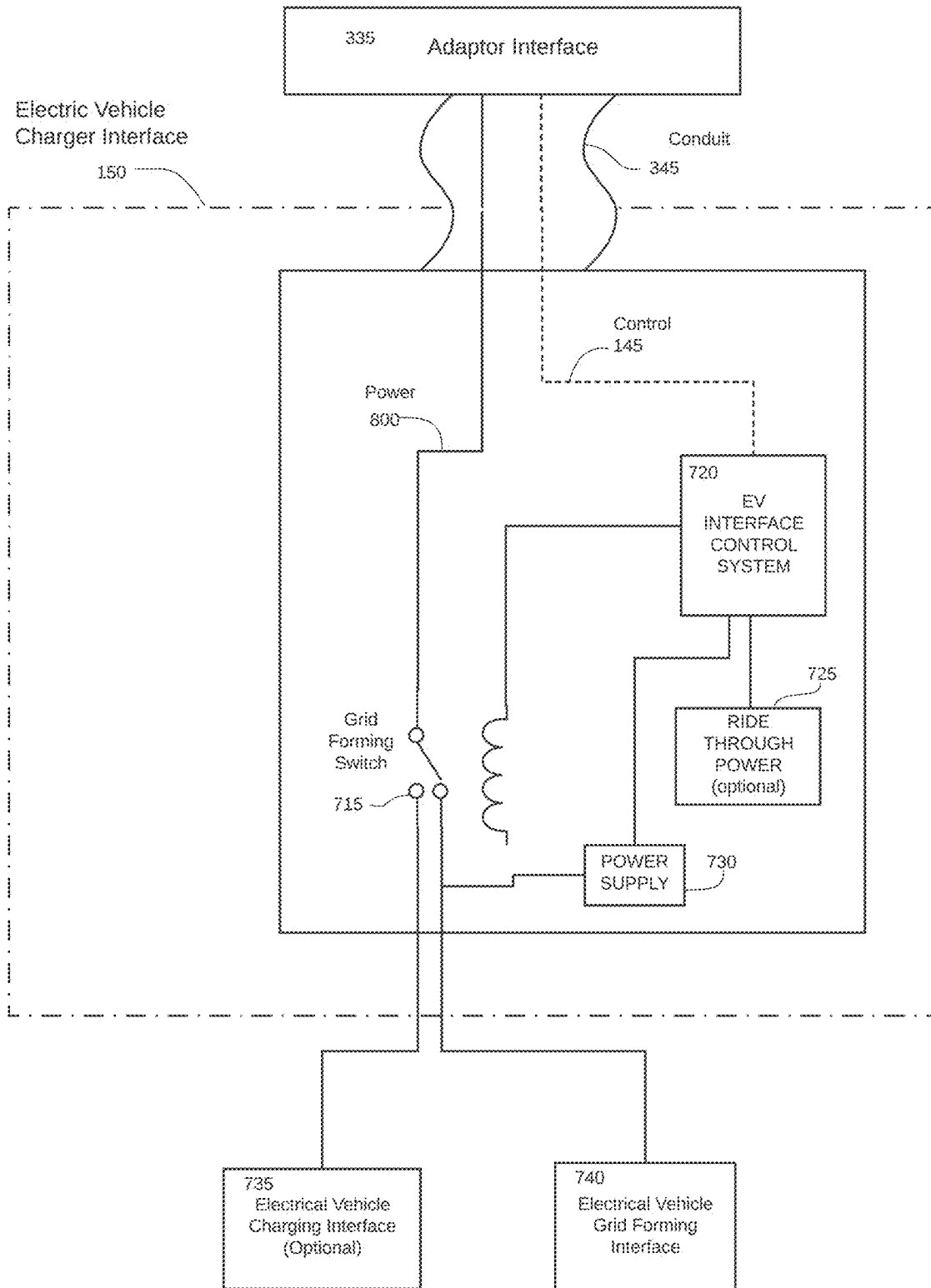
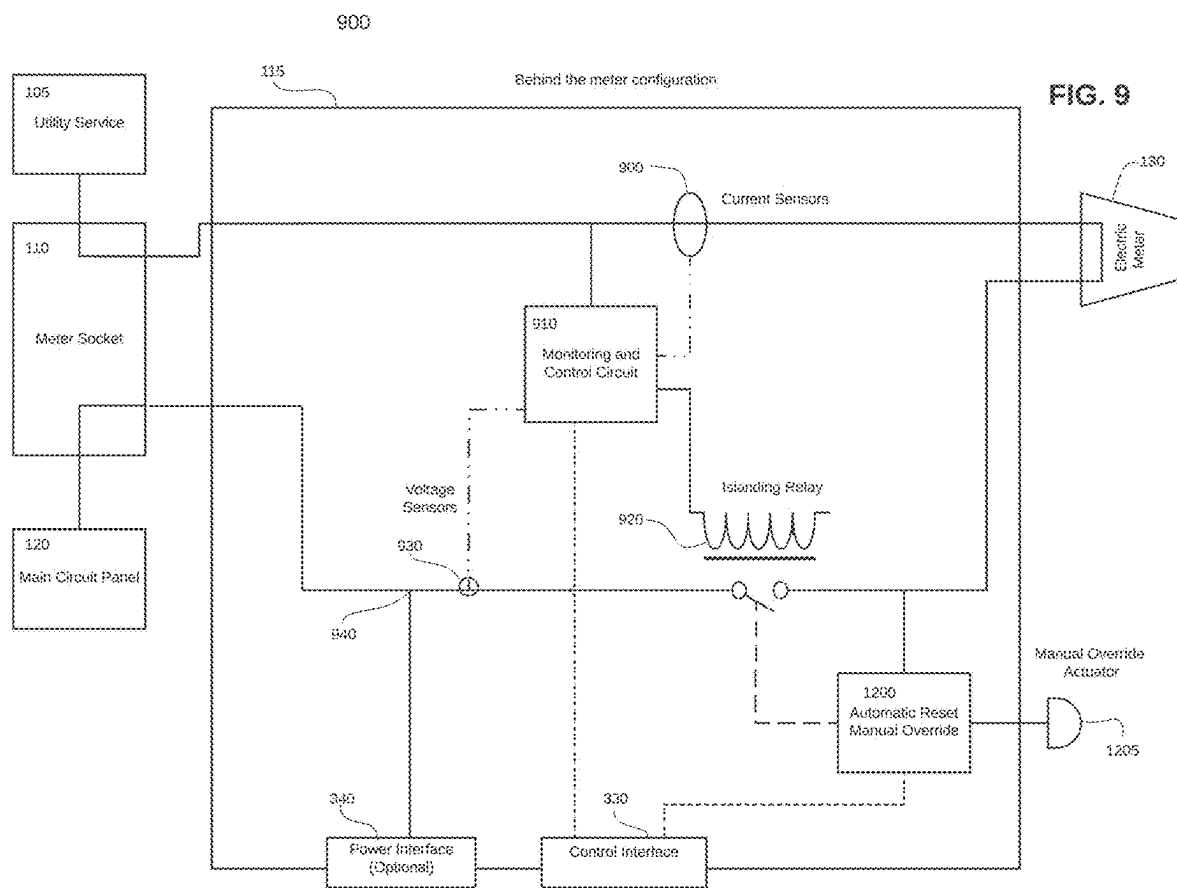
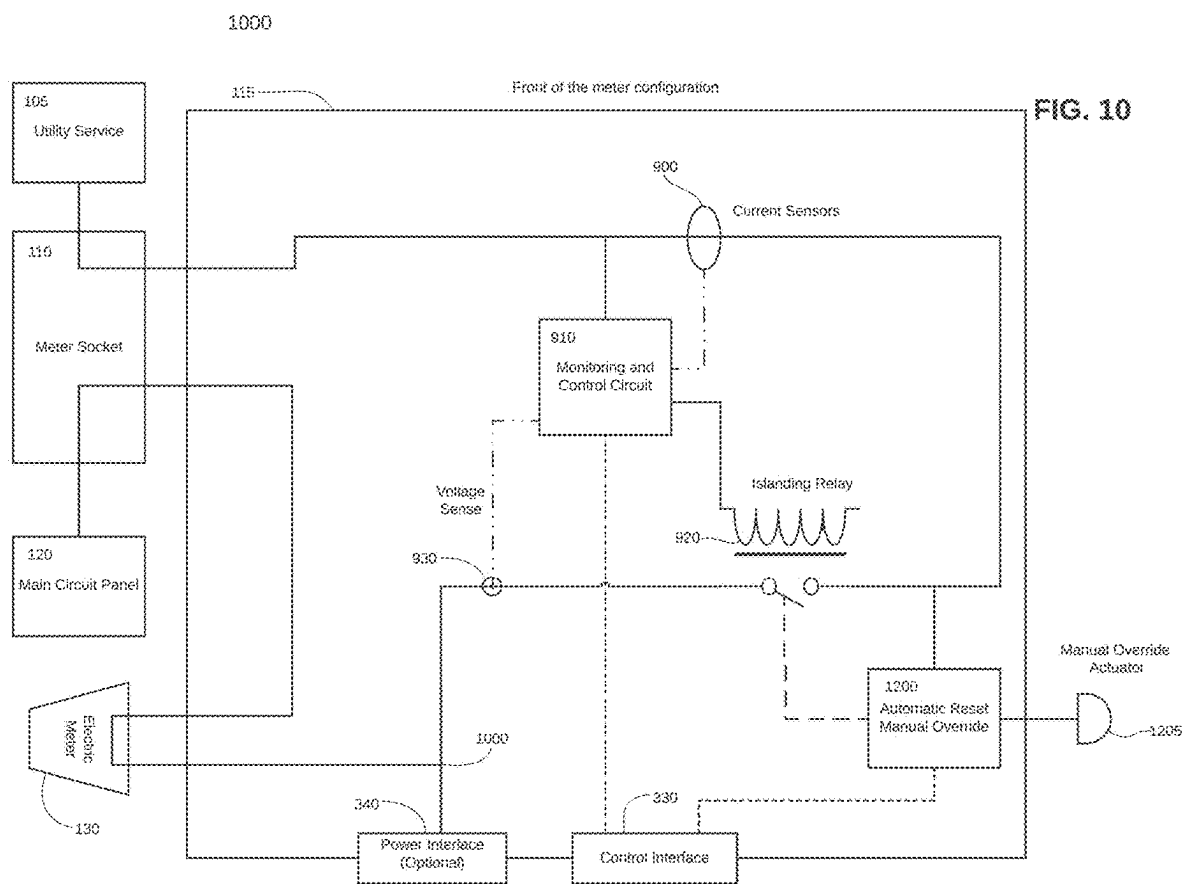


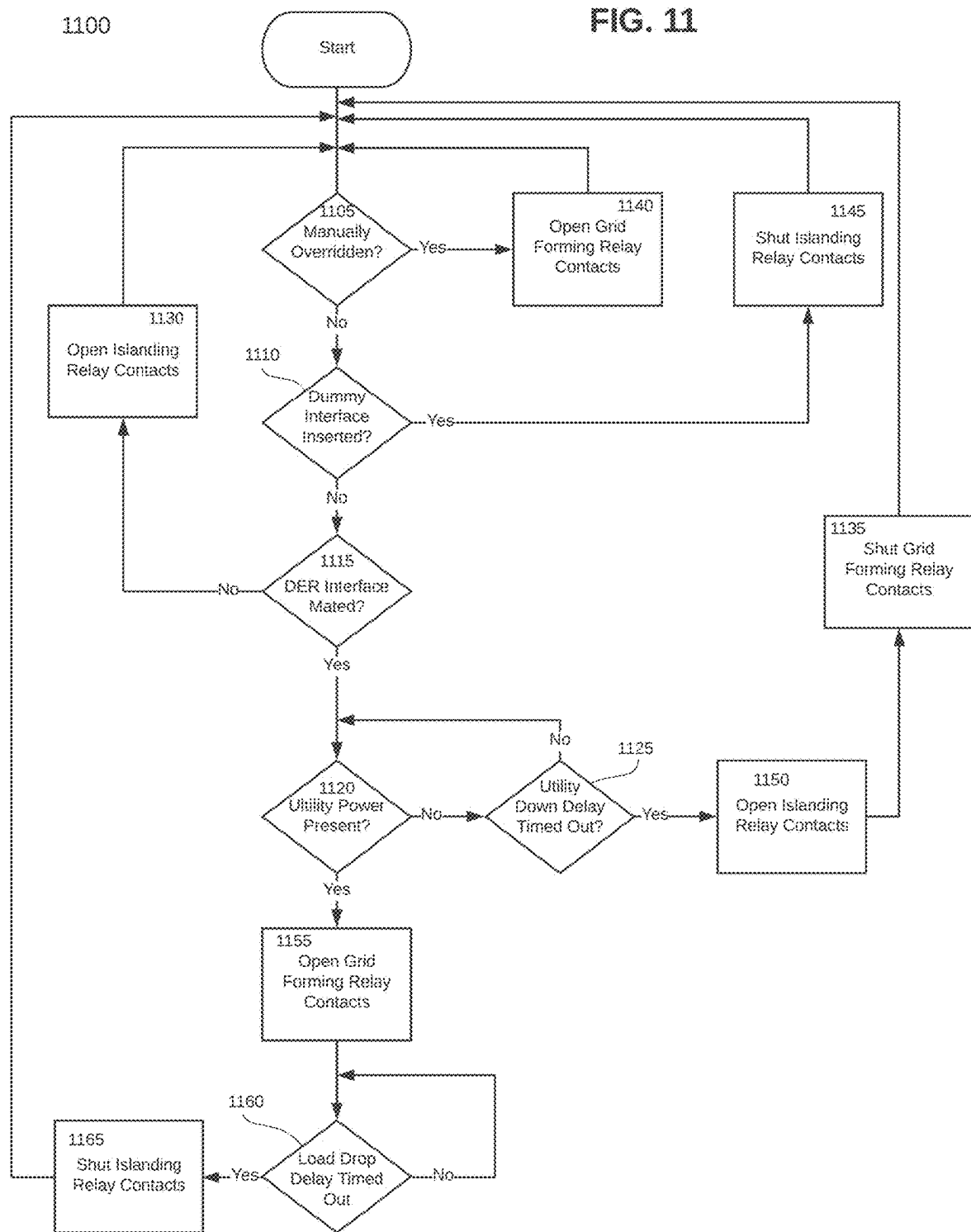
FIG. 8

700









1200

FIG 12A

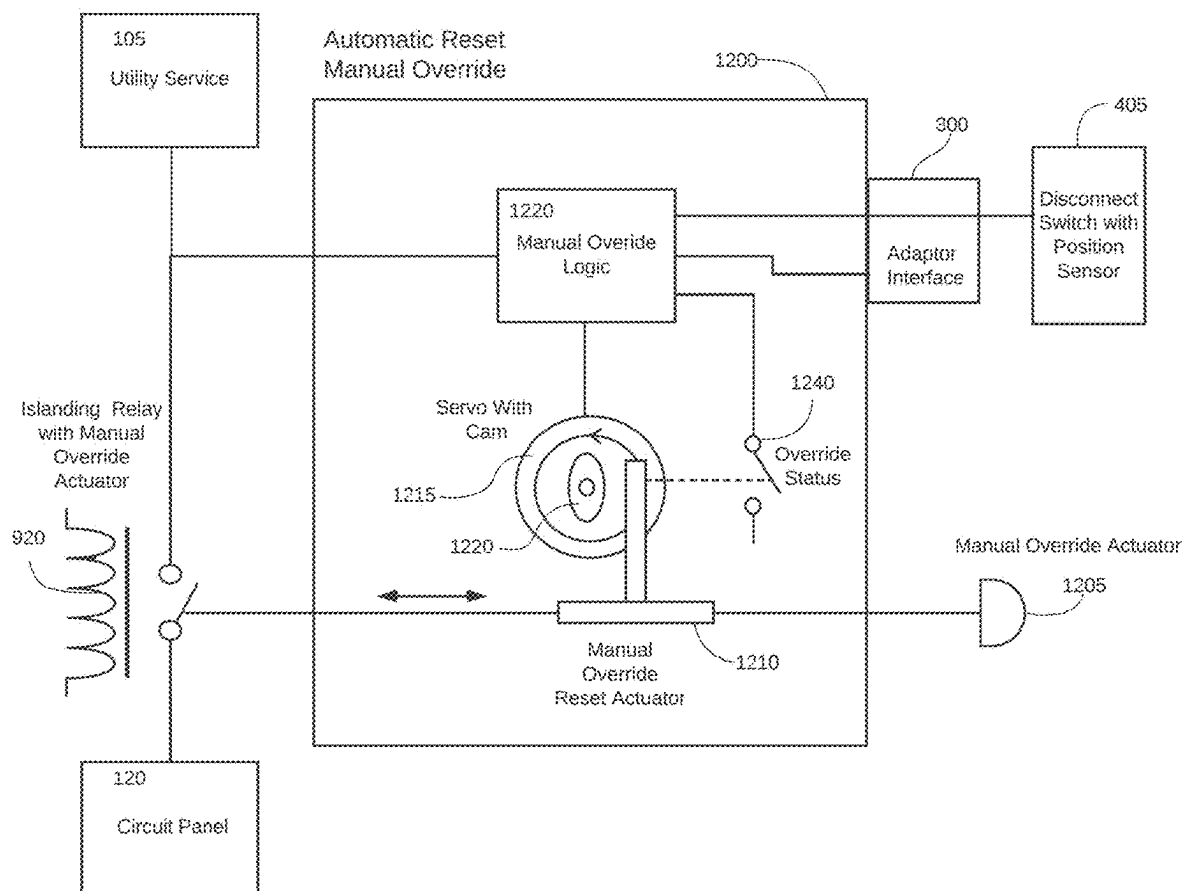


FIG. 12B

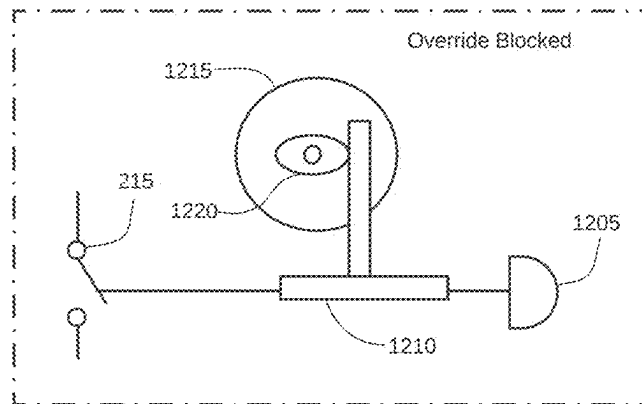
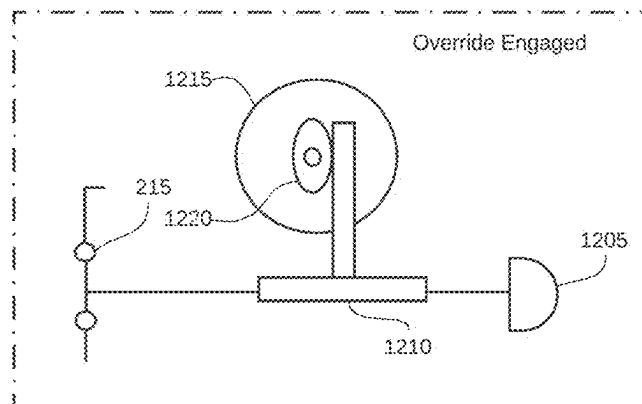
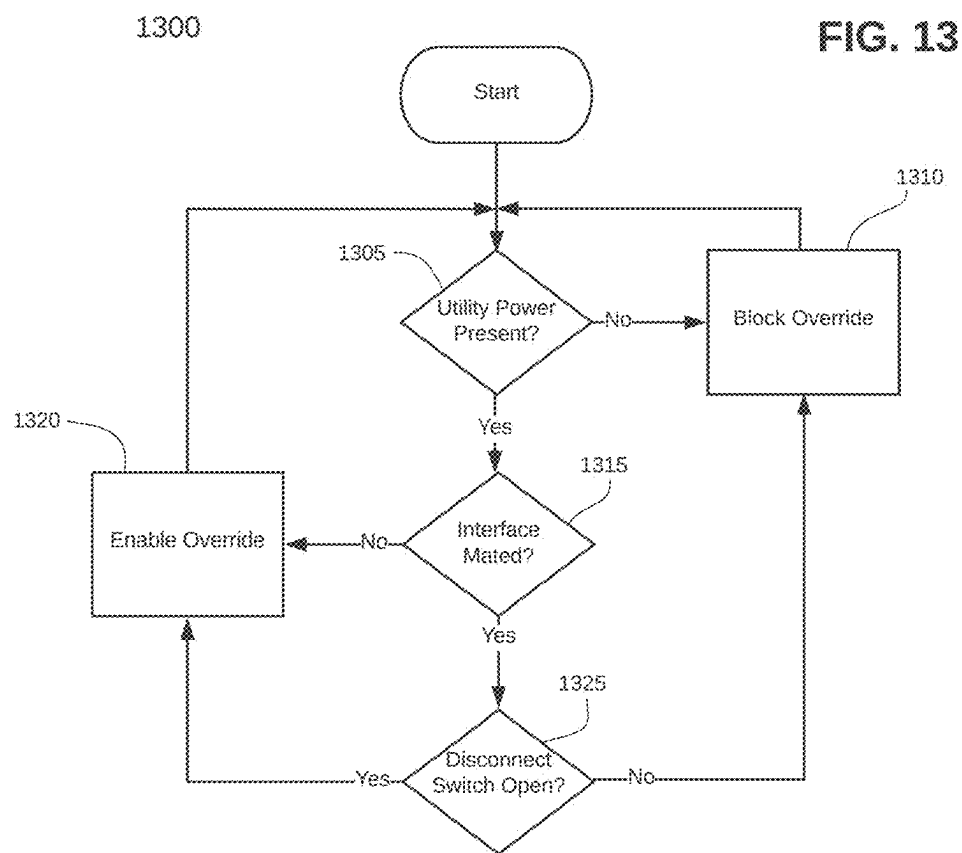


FIG. 12C





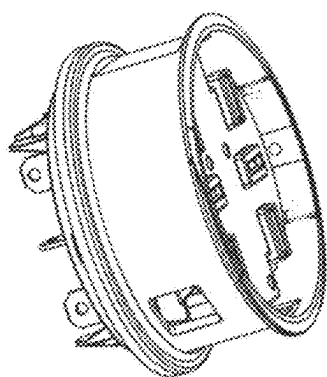


FIG. 14A

Islanding Meter Socket
Adaptor

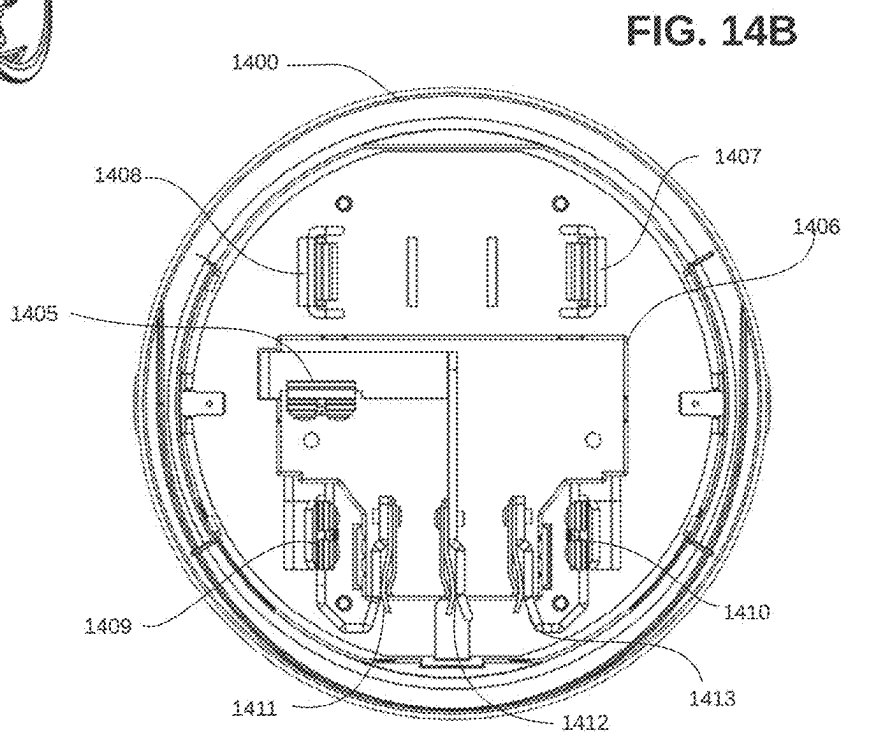
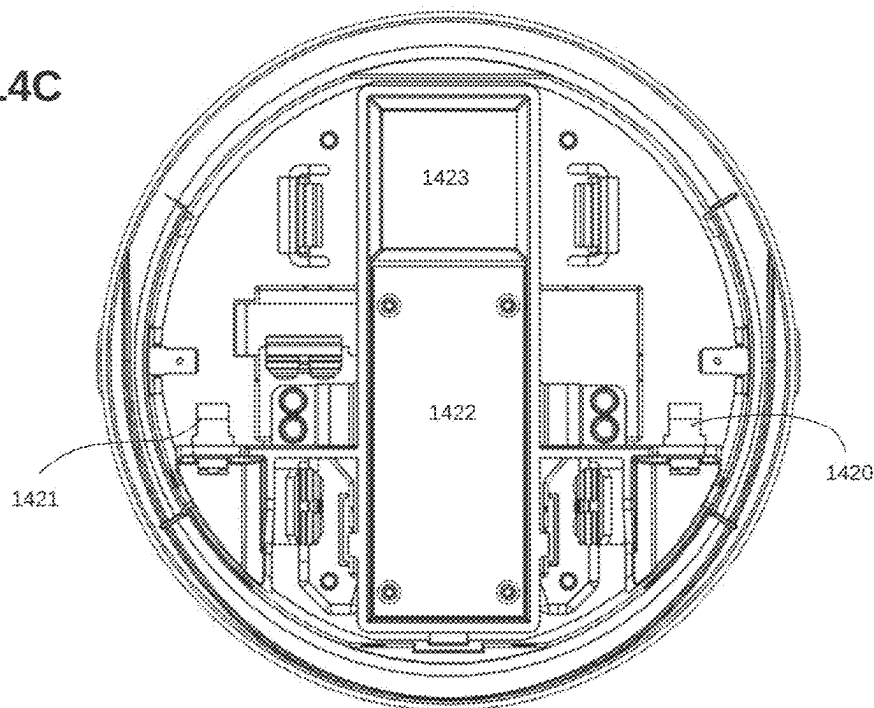


FIG. 14B

FIG. 14C



Interface Connections

FIG. 15A

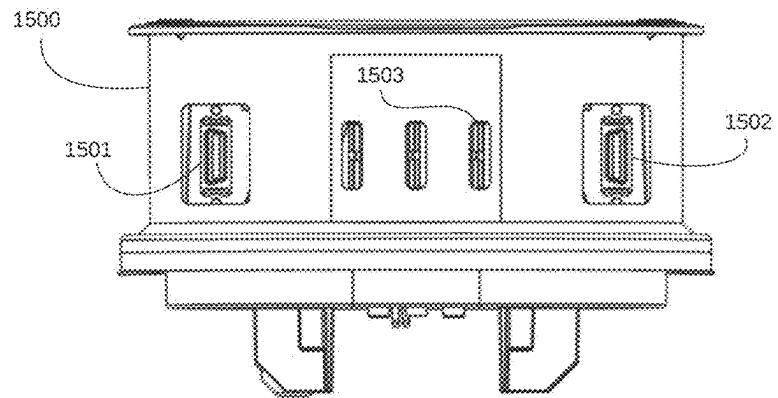
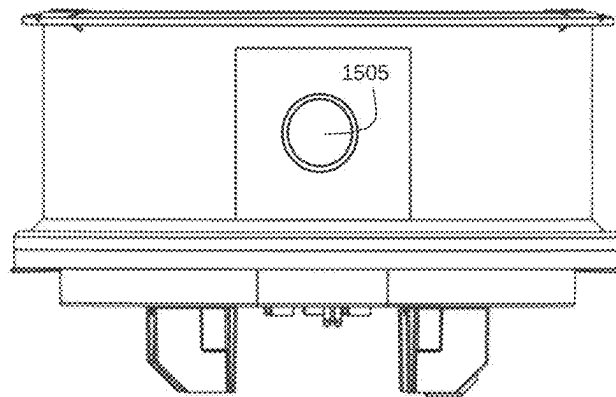


FIG. 15B



Back View

FIG. 16A

1600

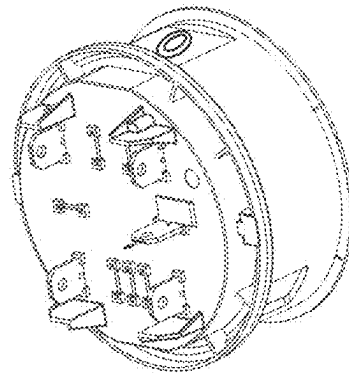
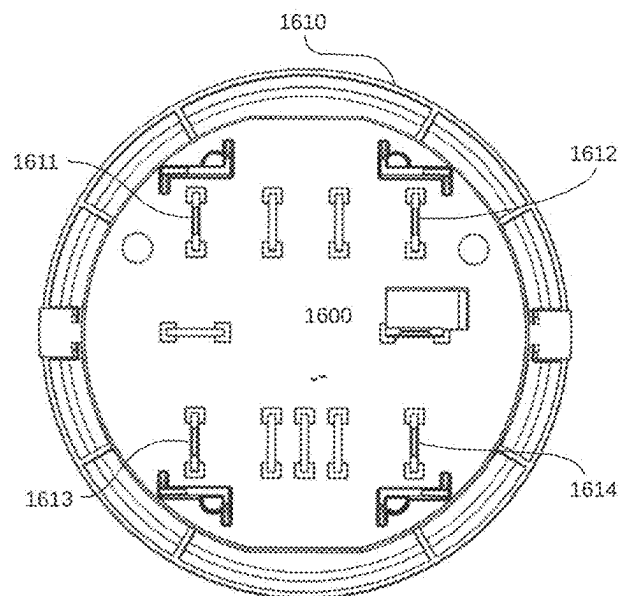


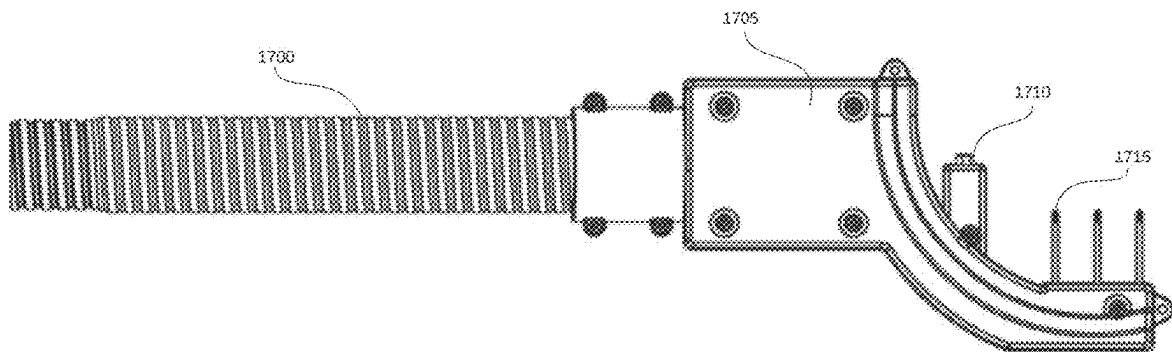
FIG. 16B

1605



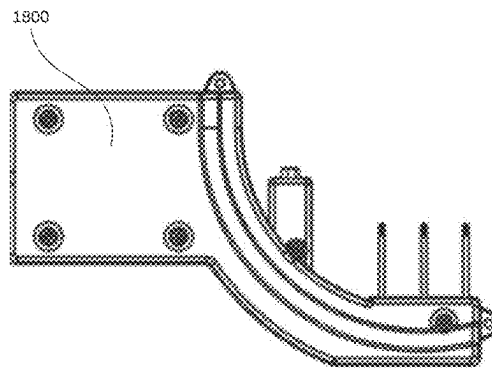
Adapter with Conduit

FIG. 17



Adapter without Conduit

FIG. 18



Circuit Breaker in Interface

FIG. 19A

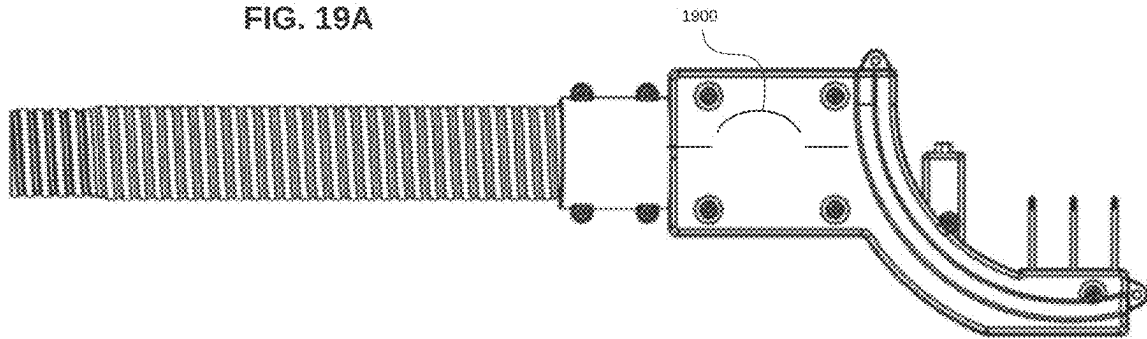
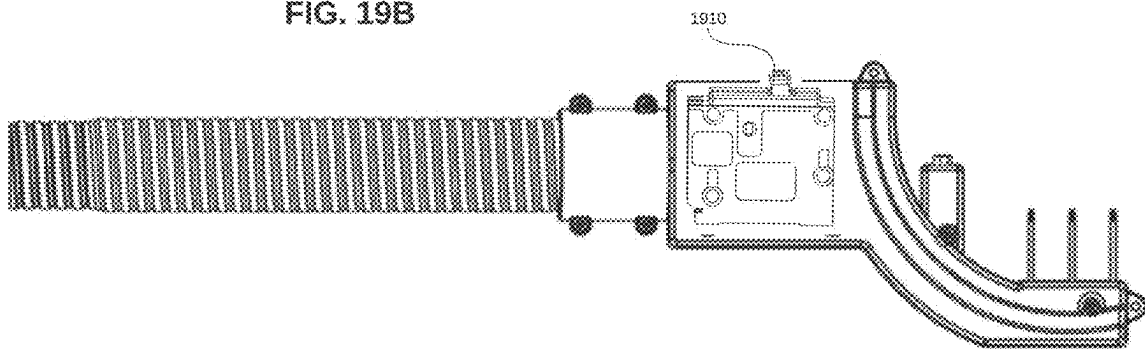


FIG. 19B



Interface Mating

FIG. 20A

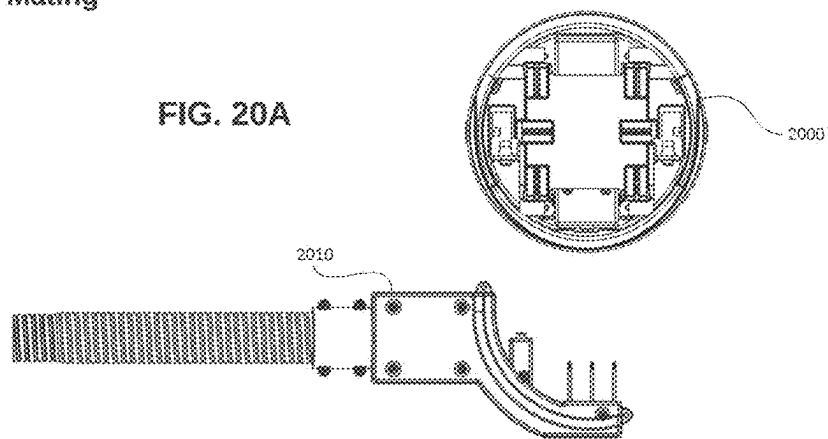
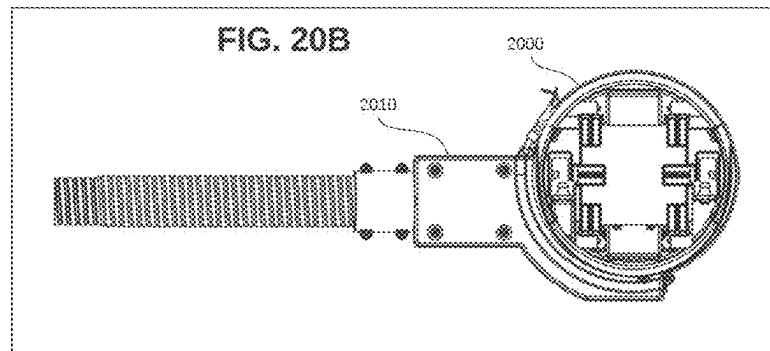


FIG. 20B



Behind The Meter Reversible
Interface

FIG. 21A

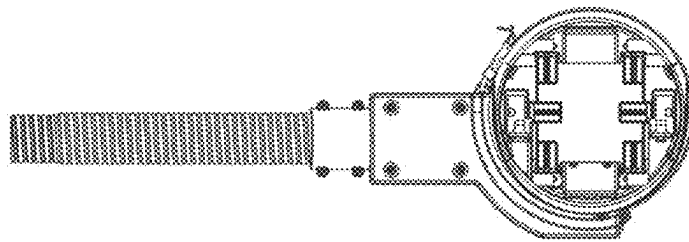
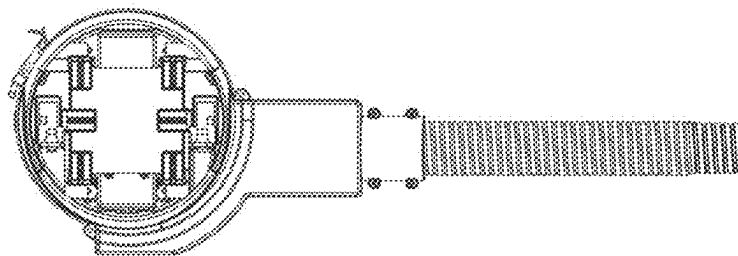


FIG. 21B



Front of The Meter Interface

FIG. 22A

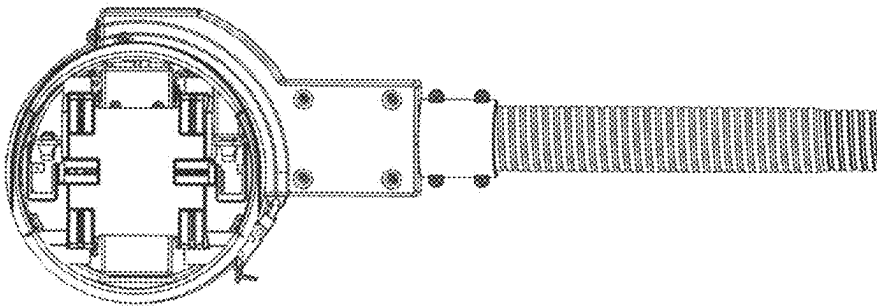
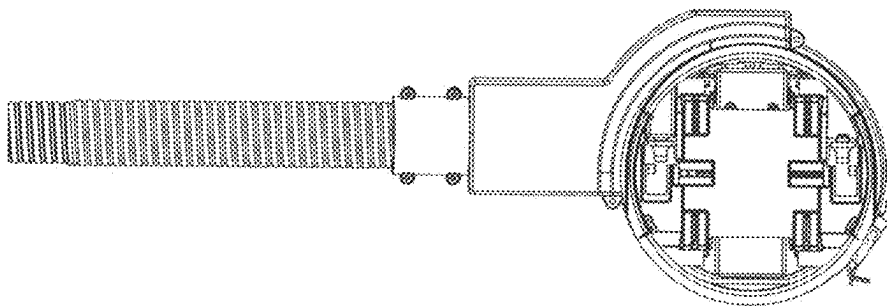


FIG. 22B



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ELECTRIC VEHICLE CHARGER WITH AUTOMATED GRID-MANAGEMENT AND VEHICLE-TO-HOME ISOLATION/BACKUP

TECHNICAL FIELD

The embodiments described and recited herein pertain, generally, to electrical power storage and deployment.

BACKGROUND

Electric vehicles (EVs) utilize battery packs, which include electrical energy storage systems that provide energy to power the vehicle and its accessories. The energy storage system corresponding to some EVs may be utilized for other purposes external to the EV itself, including powering small loads from appliances, powering site-level loads, or even returning power to the utility electric grid.

More broadly, electrical energy storage, whether housed in an EV or not, is falling in cost and increasing in prevalence. It can provide grid stability, generation cost reduction, and, when installed on-site with electrical loads, can keep those loads powered during grid outages.

SUMMARY

In accordance with another example embodiment, a consumer-side power distributor includes a first interface to bi-directionally electrically interface with a public utility-facing meter socket and a second interface to bi-directionally electrically interface with an electric vehicle charger.

In accordance with another example embodiment, a system includes utility-side power source, a customer-side power distribution system, and a meter socket to interface that serves as an electrical interface for the utility-side power source and the customer-side power distribution system. The customer-side power distribution system includes an electric vehicle charger and an islanding meter socket adaptor to electrically facilitate a bi-directional interface between the electric vehicle charger and the utility-side power source, via the meter socket.

In accordance with yet another example embodiment, a meter collar adaptor provides an interconnection point between the electric grid and a site's electric network for an energy storage system housed in an EV or otherwise weatherized for outdoor siting. The adaptor furthermore contains a disconnect switch that opens under certain conditions to isolate the site and the energy storage system below the circuit from the grid, and that is manually resettable so long as other safety conditions are met. An onboard computing system within the adaptor contains measurement of the power and related aspects flowing through the adaptor, data processing, data communications, and a data connection to the energy storage system to manage its operation in response to signals from the computing system. An associated EV charger provides a user interface to set operating modes for the energy storage system.

BRIEF DESCRIPTION OF THE DRAWINGS

In the detailed description that follows, embodiments are described as illustrations only since various changes and modifications will become apparent to those skilled in the art from the following detailed description. The use of the same reference numbers in different FIG.s indicates similar or identical items.

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FIG. 1 shows a schematic drawing of a power distribution system, in accordance with at least one example embodiment described and recited herein;

FIG. 2 shows a schematic drawing of a power distribution system, in accordance with at least one example embodiment described and recited herein;

FIG. 3 shows a schematic drawing of an energy storage system, in accordance with at least one example embodiment described and recited herein;

FIG. 4 shows a schematic drawings of an energy storage system, in accordance with at least one example embodiment described and recited herein;

FIG. 5 shows a schematic drawings of an electrical path corresponding at least one example embodiment of a weatherized energy storage system as described and recited herein;

FIG. 6 shows a schematic drawings of a wireless communications path corresponding at least one example embodiment of an energy storage system as described and recited herein;

FIG. 7 shows a schematic drawing of the electric vehicle charger interface, in accordance with at least one example embodiment described and recited herein;

FIG. 8 shows a single line schematic drawing of the electric vehicle charger interface, in accordance with at least one other example embodiment described and recited herein;

FIG. 9 shows a schematic drawing of a socket adaptor in accordance with at least one example embodiment described and recited herein;

FIG. 10 shows a schematic drawing of a socket adaptor in accordance with at least one other example embodiment described and recited herein;

FIG. 11 shows an operation flow, in accordance with at least one example embodiment described and recited herein;

FIG. 12A shows a schematic diagram of an override component, in accordance with at least one example embodiment described and recited herein;

FIG. 12B shows a schematic diagram of the override reset actuator of FIG. 12A, in accordance with at least one example embodiment described and recited herein;

FIG. 12C also shows a schematic diagram of the override reset actuator of FIG. 12A, in accordance with at least one example embodiment described and recited herein;

FIG. 13 shows an operational flow for implementing an override, in accordance with at least one example embodiment described and recited herein;

FIG. 14A shows a side view of an adaptor, in accordance with at least one example embodiment described and recited herein;

FIG. 14B shows a front view of an adaptor, in accordance with at least one example embodiment described and recited herein;

FIG. 14C shows a review of an adaptor, in accordance with at least one example embodiment described and recited herein;

FIG. 15A shows a top view of an adaptor interface, in accordance with at least one example embodiment described and recited herein;

FIG. 15B shows a bottom view of an adaptor interface, in accordance with at least one example embodiment described and recited herein;

FIG. 16A shows a side view of a socket adaptor, in accordance with at least one example embodiment described and recited herein;

FIG. 16B shows a planar view of a socket adaptor, in accordance with at least one example embodiment described and recited herein;

FIG. 17 shows a side view of an adaptor interface, in accordance with at least one example embodiment described and recited herein;

FIG. 18 shows a wireless adaptor interface, in accordance with at least one example embodiment described and recited herein;

FIG. 19A shows a side view of an adaptor interface, in accordance with at least one other example embodiment described and recited herein;

FIG. 19B shows components of the adaptor interface of FIG. 19A, in accordance with at least one example embodiment described and recited herein;

FIG. 20A shows a mating of a socket adaptor and adaptor interface, in accordance with at least one example embodiment described and recited herein;

FIG. 20B shows the mated socket adaptor and adaptor interface of FIG. 20A, in accordance with at least one example embodiment described and recited herein;

FIG. 21A shows a configuration of the adaptor interface, in accordance with at least one example embodiment described and recited herein;

FIG. 21B shows another configuration of the adaptor interface, in accordance with at least one example embodiment described and recited herein;

FIG. 22A shows another configuration of an adaptor interface, in accordance with at least one example embodiment described and recited herein; and

FIG. 22B shows another configuration of the adaptor interface of FIG. 22A, in accordance with at least one example embodiment described and recited herein.

DETAILED DESCRIPTION

In the following detailed description, reference is made to the accompanying drawings, which form a part of the description. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. Furthermore, unless otherwise noted, the description of each successive drawing may reference features from one or more of the previous drawings to provide clearer context and a more substantive explanation of the current example embodiment. Still, the example embodiments described in the detailed description, drawings, and claims are not intended to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein and illustrated in the drawings, may be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are explicitly contemplated herein.

Described and recited herein are embodiments of an electric vehicle charger and related energy storage devices having grid-management capabilities and on-site powering capabilities. The present embodiments, which are not intended to be limiting, present cost-effective solutions while adhering to regulated standards, e.g., National Electrical Code rules. Further, the embodiments described and recited herein are weather agnostic, i.e., deployment and operation thereof are not dependent upon moderate temperature ranges and do not require installation in a weatherized environment. Further still, the embodiments described and recited herein include dedicated circuits and/or switch gear to implement ride-through of a site-level power outage.

An example embodiment of the invention includes a complete electric vehicle or related battery storage system comprised of battery and power electronics housing and meter adaptor or meter connected by a pluggable interface. In addition, batteries, power electronics for control, current sensing, local area communication and utility communications may reside in the battery housing. Utility communications use either cellular or advanced metering infrastructure for communications. A meter or plastic meter collar containing grid voltage sensing, line-side disconnect, connection for plug terminals, and whole-house current sensing. Control software for battery power with multiple settings including—grid support, islanded home back up, electric vehicle support. Software makes decisions from circuit specific requirements due to each site's specific address and connection to grid. An onboard computing platform can be used to make local autonomous decisions about the best mode of operations, either in isolation or in coordination with other systems. A learning algorithm may be implemented to increase the efficiency of the operational decision making.

FIG. 1 shows a schematic drawing of a power distribution system, in accordance with at least one example embodiment described and recited herein. As depicted, system 100 includes, at least, utility service 105, meter socket 110, islanding meter socket adaptor 115, service panel 120, loads 125, electric meter 130, electric vehicle charger interface 150, and electric vehicle 155.

Utility service 105 may refer to a public utility distribution system that supplies two-phase power via a meter to a customer.

Meter socket 110 may refer to a socket having utility-side contacts to interface with utility service 105 and customer-side contacts to electrically connect to a customer's private distribution system, e.g., islanding meter socket adaptor 115 and service panel 120.

Islanding meter socket adaptor 115 may refer to a device depicted in FIG. 14A that plugs into the meter socket 110 and resides between the meter socket and electric meter 130.

Service panel 120 may refer to the main service panel that provides a distribution point for the customer's loads 125. This panel usually contains circuit protection devices and terminal points that enable the distribution of power through the private facility.

Loads 125 may refer to the typical loads found in a private facility. Some examples of typical loads include, but not limited to, electric appliances, air conditioners, electric hot water heaters, and electric outlets.

Electric meter 130 may refer to a power metering device that plugs into the Islanding meter socket adaptor 115. Electric meters are typically utilized to monitor the power consumption or production at a Utility customer's facility for the purpose of billing for the utility service.

Electric vehicle charger interface 150 may refer to a charger interface that may interact with islanding meter socket adaptor 115 to safely enable an electric vehicle 155 to connect to the load side of meter socket 110. In accordance with at least one example embodiment described and recited herein, electric vehicle charger interface 150 resides onboard electric vehicle 155 to reduce the number of external enclosures needed for system 100. Alternative form factors include separate enclosures that house EV car chargers or other multi source backup inverters.

In accordance with one example operational flow, as described and recited herein, power may be supplied to electric vehicle charger interface 150, via utility service 105, meter socket 110, islanding meter socket adaptor 115, and

wires 140; and communications in the form of digital or analog signals from, e.g., monitoring and control circuit 910 (see FIG. 9), to enable or disable the connection of the electric vehicle backup source 155 may supplied be to electric vehicle charger interface 150 via wires 145. In addition, control interface 145 may communicate with the electric vehicle inverter 155 to modify the inverters parameters including, but not limited to, power output level or power factor compensation.

In accordance with at least one example implementation, power may be routed directly from load 125 side of islanding meter socket adaptor 115 to electric vehicle charger interface 150. In accordance with at least one additional or alternative implementation, electric vehicle 155 may supply power to the grid or to loads 125. The aforementioned implementations may be enacted depending on program settings, determined by the local utility regulations and end user inputs, programed into the monitoring and control circuit (see FIG. 9).

In accordance with a non-limiting example implementation, in the event of a grid outage, islanding meter socket adaptor 115 may be configured to disconnect from the grid, i.e., utility service 100, while maintaining an electrical connection to electric vehicle charger interface 150 as well as electric vehicle 155 and to service panel 120, thereby supplying power from electric vehicle 155 to loads 125. Islanding meter socket adaptor 115 may be configured to inform the electrical vehicle charger interface 150 that the islanding relay contacts are open by sending a signal over the control interface 145. Accordingly, electrical vehicle charger interface 150 may enable the grid forming power 740 supply by switching the grid forming switch 715 to the electric vehicle grid forming interface 740 position to energize the power connection and pick up the service panel loads 125.

FIG. 2 shows a schematic drawing of a power distribution system, in accordance with at least one other example embodiment described and recited herein. As depicted, similar to system 100, system 200 includes, at least, utility service 105, meter socket 110, islanding meter socket adaptor 115, service panel 120, loads 125, electric meter 130, electric vehicle charger interface 150, and electric vehicle 155.

In accordance with at least one example embodiment, power lines 200 electrically connect charger interface 150 and service panel 120. In at least one example implementation, charger interface supports charging of electric vehicle 155 as well as, in accordance with at least one additional or alternative implementation, discharging of electric vehicle 155 to supply power to loads 125 or to feed power back to utility service 105, i.e., the grid. That is, in the event of a power outage, a relay for meter socket adaptor may open to disconnect all downstream components from the grid, thus enabling EV 155 to provide backup power to loads 125.

FIG. 3 shows a schematic drawing of an energy storage system, in accordance with at least one example embodiment described and recited herein. That is, FIG. 3 shows islanding meter socket adaptor 115 interconnected at a site with electrical loads on a grid side circuit interconnection point, with power flowing between the grid, the site, and the energy storage system.

System 300 includes islanding meter socket adaptor 115 electrically connected to electric vehicle charger interface 150. As depicted, islanding meter socket adaptor 115 includes, at least, socket adaptor collar 305, control interface 330, adaptor interface 335, power interface 340, and conduit 345.

Adaptor interface 335 houses the mating connectors for the power and control interfaces which may be electrically connected to socket adaptor collar 305, via power interface 340, which may have plug type connectors for, e.g., AC power Line 1 (315), Line 2 (325), and Neutral (320).

Socket adaptor collar 305 may be implemented as a meter socket adaptor depicted in FIG. 14A. The meter socket adaptor houses the islanding relay 920 and, when mounted in meter socket 110, facilitates its insertion between the utility service 100 and the service panel 120 and further provides a location for the backup power supply connection facilitated by the mating of the adaptor interface 335 and power interface 340.

Control interface 330 may be integrated with the islanding meter socket adaptor 115 collar and houses the connectors used to route the control 10 signals from the islanding meter socket adaptor 115 collar to the adaptor interface 335 through the connections 310. This communication is used to control the grid forming switch 715. In addition, the communications may control EV backup source 155 to vary its output based upon desired system setpoints. The communications may be implemented in many forms including but not limited to digital IO, analog IO, and industrial standard serial protocols.

Power interface 340 may be configured to house pluggable connectors 305 designed to pass backup power to the socket adaptor collar 305. These connectors are appropriately sized to pass voltage and current required for the backup power level. Typical electric vehicle backup sources could provide up to 24,000 Watts or more of power to the private facility. For a 240 Volt service this would require connectors capable of carrying up to 100 amps of current.

Conduit 345 may be provided to facilitate AC power connectivity between islanding meter socket adaptor 115 and electric vehicle charger interface 150.

FIG. 4 shows a schematic drawing of an alternative embodiment of 300, in accordance with at least one other example embodiment described and recited herein.

As depicted in FIG. 4, system 300 further includes circuit breakers 400 and 405 that are integrated into the adaptor interface 335 providing circuit protection. Since socket adaptor 115 facilitates a power connection directly to utility service entrance at the meter socket 110, circuit protection is provided between the service entrance connection 110 and the electric vehicle backup source 155 as close as practical to the service entrance at the meter socket 110. The adaptor interface 335 is a convenient location to place this circuit protection since it is close to the service entrance and it can be easily replaced if needed. The circuit protection can consist of, but is not limited to, circuit breaks or fuses rated for the power flow being provided by the backup source.

FIG. 5 shows a schematic drawings of an electrical path corresponding at least one example embodiment of system 300, as described and recited herein.

As depicted in FIG. 5, system 300 includes islanding meter socket adaptor 115 electrically connected to electric vehicle charger interface 150. As depicted, islanding meter socket adaptor 115 includes, at least, socket adaptor collar 305, control interface 330, adaptor interface 335, power interface 340, and conduit 345.

Control interface 310 may facilitate an electrical connection between islanding meter socket adaptor 305 and electrical vehicle charger communication interface 335. Control interface 310 is the connection point for IO signals that inform electrical vehicle charger interface 150 of the islanding relay status and interactively communicate with the EV backup supply 155. This connection may be made with a

multi-pin low power signaling connector. Low power electrical signals are typically below the thresholds for limited power circuits as defined by the National Electric Code (NEC).

Wired control connection **500** is one method for establishing a signaling connection between electric vehicle charger communication interface **335** and electric vehicle charger interface **150**. Wired control connection **500** may carry **10** signals and communication traffic between the monitoring and control circuitry **910** to electrical vehicle charger interface **150**.

FIG. **6** shows a Schematic drawing of an alternative embodiment of system **300**, as described and recited herein.

As depicted in FIG. **6**, system **300** further includes wireless communications path **610** to **600** between electric vehicle charger communication interface **335** and electric vehicle charger interface **150**. Further, control interface plug **310** may be provided to electrically connect islanding meter socket adaptor **305** and electric vehicle charger communication interface **335**. Wireless communications protocols implemented between wireless interfaces **600** and **610** may include, but not be limited to, Wi-Fi, PLC, BPL, etc.

FIG. **7** shows a schematic drawing of the electric vehicle charger interface, in accordance with at least one example embodiment described and recited herein.

Electric vehicle charger interface **150** may be configured or designed to switch between normal grid-based vehicle charging and electrical vehicle-to-grid, or other backup power sourcing, protocols. Electrical vehicle charger interface **150** may reside in a separate enclosure that provides the interconnection between islanding meter socket adaptor **115** and the electric vehicle. The electrical vehicle charger interface houses the components needed to transfer the power connection to the grid forming interface in the event of a utility power outage.

As depicted in FIG. **7**, electric vehicle charger interface **150** may include fused disconnect switch **705**, grid forming switch enclosure **701**, which houses the grid forming switch **715**, electric vehicle interface control system **720**, ride-through power **725**, power supply **730**. The electric vehicle charger interface provides terminal connections for the electric vehicle charging interface **735**, and electric vehicle grid-forming interface **740**.

Based on data regarding grid power status received via adaptor interface **115**, electric vehicle charger interface **150** may operate internal grid forming switch **715** to charge electric vehicle **155**; alternatively, based on the data regarding grid power status received via adaptor interface **115**, electric vehicle charger interface **150** may operate internal grid forming switch **715** to switch to islanding mode to for devices capable of operating in islanding mode.

As referenced in the description and recitation herein, islanding may refer to the disconnection of the service panel **120** from a utility service such that the service panel **120** may be energized from EV backup source **155** through the islanding meter socket adaptor **115** without back feeding power onto the utility service.

Electric vehicle charger interface **150** may be powered either by power supply **730**, which may be electrically connected to the electric vehicle grid forming interface **740**, or by ride through power **725** located in the electric vehicle charger interface. Ride through power **725** may be included with the electric vehicle charger interface **150** to power the EV Interface Control system in the event that there is no utility or backup power and it is desired to keep the control system powered for status communications or indications. In this regard, vehicle charger interface electric vehicle charger

interface **735** also includes fused disconnect switch **705**, which provides circuit protection for the service entrance conductors **106** and additionally gives a visible disconnect for service personnel to open in for system maintenance.

FIG. **8** shows schematic drawing of an alternative embodiment of electric vehicle charger interface **700**, in accordance with at least one other example embodiment described and recited herein.

As depicted in FIG. **8**, system **700** does not include a fused disconnect switch. Accordingly, the fusing means would be integrated into the Adaptor Interface as shown in FIG. **4**. By incorporating the circuit protection in adaptor interface **335**, the fusible disconnect is no longer needed, thus reducing a number of components in the system.

FIG. **9** shows a schematic drawing of a socket adaptor in accordance with at least one example embodiment described and recited herein.

FIG. **9** depicts what may be referred to, as a non-limiting example, an islanding meter socket adaptor (IMSA), in accordance with a “behind the meter” (BTM) configuration.

As referenced herein, behind the meter refers to a configuration by which islanding relay **920** is located on a common electric side of electric meter **130** as main service panel **120**. Accordingly, main service panel **120** may be electrically isolated from utility service **105**, and thus power may be supplied through power interface **340** to supply loads wired into the main service panel **120**. While islanding relay **920** may be open, power supplied through power interface **340** is not metered by electric meter **130**. In this configuration, the backup source is usually provided by the facility owner and the generated power should not be recorded or billed by the utility. In addition, the backup power will not energize utility service **105**, ensuring safety of maintenance personnel that may be handling components at utility service **105** during a utility power outage.

Islanding meter socket adaptor (hereafter “IMSA”) **115** may couple electrically with meter socket **110** and also couple electrically with electric meter **130**. Utility power may flow from utility service **105** to the meter socket **110**, to IMSA **115**, through electric meter **130**, through islanding relay **920**, to meter socket **110**, and ultimately to the service panel **120** to power the connected loads. The islanding relay **920** is a latching relay that disconnects the utility power lines from the main service panel thus electrically islanding the main service panel from the utility service. Power can also flow in either direction from the adaptor interface **335** through power interface **340** to the power interconnection point **940**. If a load is connected to the power interface, power will flow out of interconnection point **940** to supply the load. For example, a car charger will consume power to charge the electric vehicle batteries. If a power source is connected to the power interface, the power can flow into connection point **940**. For example, a backup generator can source power that will flow into the connection point and supply the service panel loads.

The IMSA **115** contains a monitoring and control circuit **910** that operates in concert with other control logic accessed through the control interface **330** to operate the islanding relay **920** in accordance with the logic depicted in FIG. **11**. The control circuit’s **210** power supply is auctioneered, meaning whichever power source is energized may power the circuit, whether the utility service inlet and the control interface **225** power line. This allows the control circuit to operate when either utility power **100** or the EV grid forming source **600** is online. In addition to controlling the relay, monitoring and control circuit **910** measures current sensors **900** and voltage sensors **930** to provide input

into the control logic for operation of islanding relay **920** and information for auxiliary functions such as, but not limited to, controlling the power output of the EV backup source **155**.

If control circuit **910** fails, the islanding relay **920** has the capability to be overridden by a manual override actuator **1205** which is externally accessible on the IMSA. The IMSA also contains an Automatic Reset Manual Override (ARMO) **1200** that will open and disable the manual override of the islanding relay **920**. The ARMO **1200** only allows the islanding relay **920** to be overridden shut when both the electric vehicle power source is disconnected from the adaptor interface **335** and power is present on the utility service **100**. This is an additional safety feature to prevent the Utility service **100** from being energized through the adaptor interface **335** when there is a power outage and maintenance is being performed on the utility system. The ARMO **1200** is totally separate from the control circuit **910** and is powered by the utility line so it will not be affected by a failure in the control circuit **910**. The ARMO **1200** also provides override status to the control system for use in the control logic.

In addition to monitoring the current flowing from the Utility service **100**, the Current sensors **900** can provide circuit protection for the service entrance components. When power is being supplied to the standard main service panel **120** and to an additional load through the power interface **340** it is possible to overload the meter socket **110** and utility service **100** drop. Typically, standard service panels include a main circuit protection device that are sized to the maximum service drop rating. In the event that the main service panel **120** draws the maximum amount of power from the service drop and an additional load is then connected to the interface adaptor power interface **340**, the standard meter socket adaptor **110** is in danger of being overloaded. In this case, the current sensors **900** are used to detect the total load being serviced by the utility service **110**. If the load rating is exceeded the monitoring and control circuit **910** will either disconnect the additional load by opening the grid forming switch **715** or send a signal to the load to reduce its draw such that the total draw is below the utility service rating.

FIG. **10** shows a schematic drawing of a socket adaptor in accordance with at least one other example embodiment described and recited herein depicts the IMSA in the Front of The Meter (FTM) configuration. The components are essentially the same as the system depicted in FIG. **9** with the exception that the standard electric meter **130** is located on the other side of the islanding relay **920**. This allows the power interface **340** to be connected to the utility service **100** side of the electric meter **130**. This configuration is typically used when the power supply providing energy through the power interface **340** is considered a utility asset that can be metered through the standard electric meter **130** when the islanding relay **920** is open.

FIG. **11** shows an operation flow, in accordance with at least one example embodiment described and recited herein. The flow pertains to the resiliency logic used to operate the relays in the IMSA **115** and electric vehicle charger interface **150**. This logic enables the safe application of electric vehicle grid forming power in the event of a utility power failure. It also prevents the power interface (Optional) **340** from being energized while the islanding adaptor interface **300** is unmated. This reduces the possibility of electric shock or spark while removing or mating the adaptor interface **335**.

The logic starts **1100** when power is applied to the control circuit by either the Utility service **100** or the EV grid

forming interface **740** source. Determining if the islanding relay **920** is manually overridden **1105** is the first decision. The ability to manually override the relay is controlled by the Automatic Reset Manual Override (ARMO) **1200** that operates in accordance with the logic depicted in FIG. **13**. If the Islanding relay **920** is overridden, then the control system opens the grid forming relay **1140** contacts. If it is not overridden, then it continues to detect if a dummy adaptor interface is inserted **1110** into the IMSA **200**.

The dummy adaptor interface is used to safely cover the IMSA interface ports when an active adaptor interface **335** is not in service. If the dummy adaptor interface is inserted, then the islanding relay **920** is shut **1145** so utility power can flow to the standard main service panel **120**. If the dummy interface is not in place, then the system detects if an active adaptor interface **335** is mated **1115** as shown in FIG. **20B**.

If the adaptor interface is not mated, then the control system opens the islanding relay **920** contacts **1130** thus de-energizing the power interface **340**. If the adaptor interface **335** is mated, then the system detects if power from the utility service **105** is present **1120**. If the power from the utility service **105** is not present, then it waits for a utility down delay timed out **1125**.

The utility down delay timed out **1125** is intended to prevent rapid cycling of the relay system and EV power source in the event of short power interruptions. An example of a short interruption occurs when utility power is lost due to an intermittent short circuit that is subsequently cleared after a breaker reclosing cycle. These interruptions typically are only several seconds in duration.

If the utility power is still down after the delay time out, the control system opens the islanding relay contacts **1150** and then shuts the grid forming relay contacts **1135** thus enabling the grid forming EV source to power **135** the main service panel **120**.

If utility power is present, the control system opens the grid forming relay contacts **1155** and then waits for a short delay **1160** to allow the loads on the main panel to fully drop. Transferring load too quickly can cause large system transients induced in inductive or capacitive loads that could trip power sources or circuit breakers. Once the load drop delay has expired, the islanding relay contacts are shut **1165** and the service panel **120** is powered from the Utility service **100**.

FIG. **12A** illustrates the automatic reset manual override (ARMO) **1200** component. This device allows for the manual operation of the Islanding relay **920** contacts under certain conditions. One possible condition is where the monitoring and control circuit **910** fails and the Islanding relay **920** is in the open position. If the utility power returns to service, the manual override actuator **1205** can be utilized to shut the islanding relay **920** contacts and restore power to the service panel **120**.

The ARMO **1200** device is totally separate from the control circuit **210** so that failure in the control circuit is unlikely to affect the ARMO **1200** thus enabling power restoration if the utility power is present.

The ARMO **1200** will also automatically open and lock-out the Islanding relay **920** contacts if utility power is lost. This prevents any power generation source that is connected to the load side of the islanding relay from feeding power back to the utility grid. This protects utility maintenance personnel from being exposed to energized components during a utility power outage.

The ARMO can be reset and locked out by moving the manual override reset actuator **1210** with a cam device **1220** causing the ARMO to be disabled. The cam device **1220** is

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motivated by a servo or solenoid **1215** that is controlled by the manual override logic **1220** depicted in FIG. **15**.

FIG. **12A** shows the manual override reset actuator **1210** and servo with cam **1220** device in the position that enables shutting of the islanding relay **920** using the manual override actuator **1205**.

FIG. **12B** shows the manual override reset actuator **1210** and cam device **1220** in the position that locks out the ability to shut the islanding relay contacts **920** using the manual override actuator **1205**.

FIG. **12C** shows the manual override reset actuator **1210**, the cam device **1220**, and the islanding relay contacts **215** shut.

The override status switch **1240** indicates the position of the manual override actuator. This status can be used in the resilience logic depicted in FIG. **11**.

FIG. **13** illustrates ARMO **1200** Logic. The system starts **1300** when power is applied to the logic. The first decision is to determine if utility power is present **1305**.

If utility power is not present, the override opens the islanding relay contacts and blocks the override **1310** from being actuated. Opening the islanding contacts prevents any grid forming generator from feeding power to the grid while maintenance personnel could be exposed to energized components during a utility power outage.

If utility power is present, then it determines if an adaptor interface **335** is mated **1315** with the islanding meter socket adaptor **300**.

If the interface is not mated, then the override is enabled **1320** allowing the islanding relay to be shut. When the interface is not mated there is no path for grid forming generator power to flow back to the utility therefore it is safe to override the islanding relay. If the interface is mated, then the system determines if the disconnect switch is open **1325**.

If the disconnect switch is open, then the override is enabled **1320**. When the disconnect is open there is no path for grid forming generator power to flow back to the utility therefore it is safe to override the islanding relay.

If the disconnect is shut, the override is blocked **1310** from being actuated. Blocking the override prevents any grid forming generator from feeding power to the grid while maintenance personnel could be exposed to energized components during a utility power outage.

FIG. **14A** depicts one embodiment of the islanding meter socket adaptor.

FIG. **14B** illustrates the enclosure **1400** and main power carrying components. The meter jaws **1408**, **1407**, **1409**, **1410**, **1405**, the mate socket jaws **1411**, **1412**, **1413**, and the islanding relay **1406**.

FIG. **14C** illustrates the control electronics enclosure **1423**, electronics PCB **1422**, and control interface connectors **1420**, **1421**.

FIG. **15A** illustrates the control interfaces **1501**, **1502**, and the power interfaces **1503**.

FIG. **15B** illustrates the Automatic Reset Manual Override actuator **1505**.

FIG. **16A** depicts the back view of the meter socket adaptor this side mates with a standard meter socket.

FIG. **16B** illustrates the meter socket adaptor enclosure **1610** back interface components including the meter socket stabs **1611**, **1612**, **1613**, **1614**.

FIG. **17** illustrates the adaptor Interface **335** with a conduit **1700** to route the power and control cables to the electric vehicle charger interface **150**. The other components are the interface casing **1705**, the control interface connector **1710**, and the power interface stabs **1715**.

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FIG. **18** depicts the Adaptor Interface **335** without power or control cabling routed to the electric vehicle charger interface **150**. This embodiment shows the configuration utilizing wireless communications between the adaptor and charger.

FIGS. **19A** and **19B** depict the adaptor with a disconnect and circuit protection devices in the adaptor **1900**, **1910**. This enables the system to be deployed without a separate fusible disconnect thus reducing the number of components needed to install at the site.

FIGS. **20A** and **20B** illustrate the mating of the interface adaptor **2010** and the meter socket adaptor **2000**. FIG. **20A** shows the adaptor unmated. FIG. **20B** shows the adaptor mated.

FIGS. **21A** and **21B** show the interface adaptor mated in two different directions. The interface adaptor is symmetrical so it can be mated such that the conduit exits in the right-hand or left-hand directions. This enables installation flexibility to ease the placement of the other system components and avoid potential interferences.

FIGS. **22A** and **22B** show the meter and interface adaptors rotated 180 degrees to enable a "front of the meter" (FTM) connection. This configuration is depicted in FIG. **10**. This configuration can also mate in right-hand or left-hand configurations to facilitate installation.

From the foregoing, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

We claim:

1. A system, comprising:

an islanding meter socket adaptor that electrically facilitates a bi-directional interface between an electric vehicle charger and a utility-side power source, via a meter socket that electrically interfaces the utility-side power source to a customer-side power distribution system,

wherein the islanding meter socket adaptor includes:

a pluggable AC power interface that electrically connects the electric vehicle charger to the islanding meter socket adaptor to facilitate the bi-directional power flow between the electric vehicle charger, the utility-side power source, and/or one or more customer-side loads, via the meter socket;

an islanding relay to regulate an electrical connection between the utility-side power source and the one or more customer-side loads,

a control interface that opens the islanding relay to electrically disconnect the islanding meter socket adaptor from the utility-side power source in an islanding mode such that the electrical vehicle charger is configured to be connected to the one or more customer-side loads via the meter socket, and

wherein the control interface transmits a signal to the electrical vehicle charger that the islanding relay is open in the islanding mode such that the electrical vehicle charger operates a grid-forming switch located within the electrical vehicle charger to connect the electrical vehicle charger to be connected to the one or more customer-side loads via the meter socket.

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2. The system of claim 1, wherein the pluggable AC power interface houses connectors to pass backup power from the electric vehicle charger towards the utility-side power source.

3. The system of claim 2, wherein the connectors are electrically connected to the islanding meter socket adaptor.

4. The system of claim 1, wherein the pluggable AC power interface and the electric vehicle charger are communicatively connected wirelessly.

5. The system of claim 1, wherein the pluggable AC power interface and the electric vehicle charger are communicatively connected by a wired connection.

6. The system of claim 1, wherein the islanding meter socket adaptor includes an adaptor interface to provide electrical protection between the electric vehicle charger and utility-side power source.

7. The system of claim 6, wherein the adaptor interface includes circuit breakers.

8. The system of claim 1, wherein the islanding meter socket adaptor includes an override component to enable manual operation of the islanding relay.

9. The system of claim 1, wherein islanding meter socket adaptor includes a protection circuit that will prevent cumulative overloading of the meter socket.

10. The system of claim 1, wherein the islanding meter socket adaptor is further configured to communicate with the electric vehicle charger to modify operational parameters for an inverter connected to the electric vehicle charger.

11. An islanding meter socket adaptor configured to electrically facilitate a bi-directional interface between an electric vehicle charger and a utility-side power source, via a meter socket, the islanding meter socket adaptor comprising:

a pluggable AC power interface that is configured to receive a pluggable adaptor interface coupled to the electric vehicle charger to electrically connect the electric vehicle charger to the islanding meter socket adaptor to facilitate the bi-directional power flow between the electric vehicle charger, the utility-side power source, and/or one or more customer-side loads, via the meter socket;

an islanding relay to regulate an electrical connection between the utility-side power source and the one or more customer-side loads;

contacts configured to plug into the meter socket; and

a control interface configured to open the islanding relay to electrically disconnect the islanding meter socket adaptor from the utility-side power source in an islanding mode such that the electrical vehicle charger is configured to be connected to the one or more customer-side loads via the meter socket, and

wherein the control interface transmits a signal to the electrical vehicle charger that the islanding relay is open in the islanding mode such that the electrical vehicle charger operates a grid-forming switch located within the electrical vehicle charger to connect the

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electrical vehicle charger to be connected to the one or more customer-side loads via the meter socket.

12. The islanding meter socket adaptor of claim 11, wherein the pluggable AC power interface houses connectors to pass backup power from the electric vehicle charger towards the utility-side power source.

13. The islanding meter socket adaptor of claim 11, further comprising an adaptor interface to provide electrical protection between the electric vehicle charger and utility-side power source.

14. The islanding meter socket adaptor of claim 13, wherein the adaptor interface includes circuit breakers.

15. The islanding meter socket adaptor of claim 11, further comprising an override component to enable manual operation of the islanding relay.

16. The islanding meter socket adaptor of claim 11, wherein the control interface is further configured to communicate with the electric vehicle charger to modify operational parameters for an inverter connected to the electric vehicle charger.

17. A system, comprising:

an islanding meter socket adaptor that electrically facilitates a bi-directional interface between an electric vehicle charger and a utility-side power source, via a meter socket that electrically interfaces the utility-side power source to a customer-side power distribution system, and wherein the islanding meter socket adaptor includes:

a pluggable AC power interface that electrically connects the electric vehicle charger to the islanding meter socket adaptor to facilitate the bi-directional power flow between the electric vehicle charger, the utility-side power source, and/or one or more customer-side loads, via the meter socket;

an islanding relay to regulate an electrical connection between the utility-side power source and the meter socket, and

a control interface configured to:

open the islanding relay to electrically disconnect the islanding meter socket adaptor from the utility-side power source in an islanding mode such that the electrical vehicle charger is configured to be connected to the one or more customer-side loads via the meter socket; and

transmit a signal to the electrical vehicle charger regarding the islanding mode such that the electrical vehicle charger operates a grid-forming switch located within the electrical vehicle charger to connect the electrical vehicle charger to be connected to the one or more customer-side loads via the meter socket, wherein the signal regarding the islanding mode includes the status of the islanding relay to indicate that the islanding relay is open.

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